

Abhitej Divi

<https://github.com/abhitejdivi5/projects>

Email: abhitejdivi5@gmail.com

Mobile: +1-469-487-1608

PROFESSIONAL SUMMARY

Analog IC Designer experienced in full-cycle ASIC tapeout, including design and verification of various ADCs, amplifiers, bandgap references, and LDO's. Skilled in simulation using Cadence Virtuoso tools and seeking opportunities in the analog IC design domain.

EDUCATION

- **Prairie View A&M University** Prairie View, TX
Doctor of Philosophy in Electrical and Computer Engineering Aug 2022 – Aug 2025
- **Lamar University** Beaumont, TX
Master of Engineering in Electrical Engineering Aug 2021 – Aug 2022
- **Lakireddy Bali Reddy College of Engineering** Mylavaram, INDIA
Bachelor of Engineering in Electronics and Communication Engineering Aug 2016 – Jul 2020

WORK EXPERIENCE

- **Apple New Silicon Initiative (NSI) Graduate Research Assistant** Prairie View, TX
Prairie View A&M University Jan 2023 – Aug 2025
 - Designed ADCs and Apple-funded ASIC tapeout projects, from design specifications to final GDSII submission, ensuring schematic accuracy and design rule compliance.
 - Served as Technical Advisor for the Computer Hardware and VLSI Design Club; led workshops, mentored students, and created lab manuals for analog and digital circuits.
- **Graduate Teaching Assistant** Beaumont, TX
VLSI CAD & Signal Integrity Laboratory, Lamar University May 2022 – Jul 2022
 - Worked on soft error and soft delay mitigation using a dynamic threshold technique. Assisted with Electronics-I (3421) instruction, grading.

SKILLS

- **Key Skills:** Amplifiers, Current Mirrors, Two-Stage Amplifiers, Compensation Techniques, Common-Mode Feedback, Bandgap References, Sampling Circuits, Strong-Arm Latch Comparator, ADCs, Tapeout
- **EDA Tools:** Cadence Virtuoso (Schematic & Layout), ADE Explorer, Assura (DRC/LVS/ERC), Spectre, PVT Corner Analysis, Genus, Innovus, MATLAB, LabVIEW, HSPICE, Verilog, AutoCAD, Unix/Linux
- **Courses & Certifications:**
 - High-Speed Analog Design, Mixed-Signal IC Design (SSCD Lab, IIT Kanpur – Online Lectures)
 - Analog Circuit Design and Simulation Onboarding (vIC23.1/IC6.1.8) and Cadence RTL-to-GDSII Flow training

PROJECTS

- **Continuous Time Pipeline ADC(CTP ADC)** *On going*
 - The CTPADC implementation combines the antialiasing filter and ADC into a single unit and aiming to remove the power-hungry buffer before the ADC.
 - It is implemented using ideal blocks, achieving a 100 MHz sampling rate, and features a two-stage amplifying filter with a 10 MHz UGB and a 40 dB roll-off.
- **Fully Differential 10-bit Asynchronous SAR ADC(TSMC 180nm)** *Mar 2025*
 - Implemented a high-linearity bootstrapped switch for the sample-and-hold network and verified a 1 GHz strong-arm latch dynamic comparator for high-speed SAR ADC applications.
 - Designed a fully differential 10-bit asynchronous SAR ADC achieving 100 MHz sampling rate with monotonic switching for the capacitive DAC.
 - Achieved 10-bit ENOB and 61.8 dB SQNR from spectrum analysis of the digital output.
- **Diagnostic ASIC Tapeout (Apple NSI Tape-Out)** *May 2024*
 - Designed a diagnostic ASIC operating at 150 MHz, integrating amplifiers, digital logic, and memory units, with layouts following symmetry and common-centroid constraints.
 - Performed post-layout verification, including DRC, LVS, and ERC checks. Applied shielding, guard rings, and seal rings to minimize noise coupling.
 - Implemented a 32-pin ESD-protected pad frame; completed routing and finalized GDSII submission for fabrication.
- **Fully Differential Two-Stage Self-Zero-Canceling Compensated Op-Amp with CMFB** *Jan 2024*
 - Designed a two-stage fully differential op-amp with a telescopic first stage, using cascaded transistors for compensation, achieving a 99 dB DC gain, 63° phase margin, and 243 MHz unity gain bandwidth (UGB).
 - Implemented a fully differential two-stage op-amp with common-mode feedback (CMFB), achieving 71 dB DC gain, 63° phase margin, and 236 MHz UGB.
- **Op-Amp based Bandgap, fractional bandgap and LDO Design** *May 2023*
 - Designed a self-biased, opamp-based bandgap reference with an integrated startup mechanism generating a constant 1.25V.
 - Designed Fractional Bandgap generating constant 0.6V. Designed an LDO generating a constant 1.6V.
 - All designs verified for temperature -45°C to 125°C and supply variations.

PUBLICATIONS AND CONFERENCES

- Solar Photovoltaic Diagnostic System with Logic Verification and Integrated Circuit Design for Fabrication, *Solar, Vol. 5, No. 2, MDPI*, 2025. Divi Abhitej, and Shuza Binzaid. (Utility Patent Submitted)
- Simplifying Complex Digital Sequential Circuits by an Innovative Mixed-Signal Circuit Alternative, *Computers & Electrical Engineering, Elsevier*, 2024. Shuza Binzaid, Abhitej Divi, Md. Rokonzaman.
- Presented Apple-funded tape-out work at the CHIPS Semiconductor Summit (College Station, TX) and the NSI@HBCUs Summer 2024 Convening (Apple Park, Cupertino, CA).
- A 120-MS/s Fully Differential 10-Bit SAR ADC Employing a Low-Power Monotonic Switching DAC Architecture. (Submitted - IEEE Transactions)